

BEREZOVO, Russian Federation

WMO: 236310

Lat: **63.933N**Lon: **65.050E**Elev: **105**StdP: **14.64**Time Zone: **5.00 (E05)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-40.9	-36.1	-45.4	0.4	-40.6	-40.4	0.5	-35.8	21.4	10.2	19.7	7.5	2.6	60	0.753

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.6	80.5	67.2	76.8	65.2	73.1	63.1	69.2	77.2	67.1	74.3	64.7	71.3	7.7	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.4	97.5	73.5	64.1	89.9	71.8	61.6	82.3	69.2	33.4	77.6	31.7	74.2	29.8	71.5	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.8	19.8	17.4	-44.6	85.7	6.8	2.9	-49.5	87.7	-53.5	89.4	-57.3	91.0	-62.3	93.1		
(5)			-44.3	72.1	6.2	2.2	-48.8	73.7	-52.4	75.0	-55.9	76.3	-60.4	77.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.9	-6.7	-3.8	13.4	26.2	40.1	54.6	61.4	54.2	43.8	29.9	8.7	-1.1		
	DBStd	26.41	16.39	15.99	13.57	11.95	10.28	8.93	7.64	6.89	7.36	10.04	15.78	17.47		
	HDD50	9207	1756	1506	1135	716	343	53	5	32	213	626	1239	1583		
	HDD65	13980	2221	1926	1600	1165	772	327	164	344	635	1089	1689	2048		
	CDD50	777	0	0	0	1	36	192	358	161	28	1	0	0		
	CDD65	78	0	0	0	0	1	16	53	8	0	0	0	0		
	CDH74	645	0	0	0	0	20	145	420	58	2	0	0	0		
	CDH80	115	0	0	0	0	1	25	79	10	0	0	0	0		
	WSAvg	7.5	6.3	6.6	7.7	8.6	9.1	8.8	7.6	7.1	7.3	7.5	7.0	6.6		
	PrecAvg	20.6	1.1	0.9	0.9	1.3	1.7	2.4	2.7	2.8	2.3	2.0	1.4	1.1		
(15) Precipitation	PrecMax	28.6	2.1	2.5	1.9	2.6	4.5	7.4	6.4	6.5	4.4	4.4	3.9	2.5		
	PrecMin	15.5	0.4	0.2	0.1	0.3	0.6	0.7	0.3	0.7	0.3	0.9	0.2	0.3		
	PrecStd	3.3	0.4	0.5	0.5	0.6	0.9	1.4	1.6	1.4	1.1	0.8	0.7	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.6	31.7	43.0	58.0	77.1	83.3	85.5	80.1	69.7	55.5	37.8	32.0		
		MCWB	29.2	29.7	35.2	46.6	59.2	66.0	69.7	67.5	60.4	49.8	35.8	30.6		
	2%	DB	25.3	27.4	38.5	50.2	70.3	78.1	82.2	74.3	63.2	50.1	35.0	29.2		
		MCWB	24.4	25.7	33.1	41.5	55.5	63.8	69.1	64.7	55.9	45.8	33.6	28.2		
	5%	DB	20.8	23.5	35.0	45.7	63.8	73.8	78.6	70.0	59.0	46.1	32.0	25.8		
		MCWB	19.7	22.1	30.9	38.8	52.2	61.8	67.3	62.0	53.6	43.0	30.9	24.8		
	10%	DB	16.5	19.9	31.7	41.9	57.5	69.8	75.3	66.5	55.1	42.2	29.3	21.8		
		MCWB	15.7	18.6	28.8	36.4	48.5	59.8	65.6	60.1	51.0	40.0	28.1	20.9		
	0.4%	WB	29.6	30.4	36.3	47.3	60.2	68.5	73.3	69.8	61.6	51.2	36.5	30.6		
		MCDB	30.4	31.7	41.9	56.7	73.4	77.4	82.2	78.7	68.2	54.4	37.3	31.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	24.5	25.9	33.6	42.5	56.7	66.1	70.7	66.2	57.4	46.7	33.6	28.2		
		MCDB	25.3	27.4	37.4	48.8	68.6	75.2	78.9	72.3	61.8	49.5	35.0	29.1		
	5%	WB	19.8	22.2	31.4	39.4	53.0	63.3	68.7	63.4	54.2	43.2	31.1	24.9		
		MCDB	20.7	23.5	34.7	44.9	61.9	71.9	76.3	68.4	57.8	45.8	32.0	25.8		
	10%	WB	15.7	18.5	28.6	36.8	49.4	60.6	66.8	60.9	51.6	40.3	28.1	21.1		
		MCDB	16.6	19.8	31.7	41.3	57.8	68.5	73.9	65.5	54.9	42.1	29.3	22.0		
	5% DB	MDBR	13.3	14.9	17.7	17.1	16.4	15.9	16.6	15.6	13.1	10.3	11.9	13.2		
		MCDBR	13.2	14.2	16.3	19.8	26.2	22.6	21.7	21.6	18.4	13.5	9.7	13.4		
		MCWBR	13.0	13.0	12.5	13.1	14.6	11.5	10.7	12.3	11.9	10.0	9.4	13.0		
		MCWBR	13.1	14.1	15.9	17.8	24.2	19.2	19.0	18.9	16.5	12.5	9.9	12.9		
(35) Mean Daily Temperature Range	5% WB	MCWBR	12.9	13.0	13.0	12.4	14.1	10.9	10.0	11.7	11.6	9.9	9.7	12.5		
	taub		0.221	0.280	0.325	0.354	0.362	0.380	0.422	0.390	0.338	0.301	0.221	0.223		
		taud	1.922	2.039	2.034	2.104	2.349	2.402	2.268	2.328	2.445	2.302	1.963	1.902		
(40) Clear-Sky Solar Irradiance	Ebn at Noon		106	193	233	257	269	266	248	246	238	192	102	36		
		Edh at Noon	10	22	33	39	34	33	36	31	23	17	9	4		
	RadAvg		61	305	816	1377	1639	1691	1678	1137	639	271	81	23		
(44) All-Sky Solar Radiation	RadStd		3	23	42	145	124	118	164	109	68	35	11	1		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		N/A	N/A	N/A	+1.71	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page